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Complete Python Roadmap



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Fundamentals

- **Syntax & Structure:** Variables, Data Types, Operators
- **Control Flow:** if-else, loops, break, continue
- **Functions:** def, arguments, return values, *args & **kwargs
- **Data Structures:** Lists, Tuples, Sets, Dictionaries
- **Input/Output:** Reading user input, file operations
- **Error Handling:** try-except-finally, custom exceptions

💡 **Goal:** Build a foundation strong enough to write and debug basic Python programs.



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Intermediate Concepts

- **Modules & Packages:** import, pip, creating custom modules
- **Comprehensions:** List/Dict/Set comprehensions
- **Lambda Functions:** Map, Filter, Reduce
- **String & Date Handling:** datetime, string methods, f-strings
- **Regular Expressions (Regex):** Pattern matching and data cleaning
- **OOP Basics:** Classes, Objects,
- **Pillars of OOP:**
 - Inheritance,
 - Polymorphism,
 - Encapsulation,
 - Abstraction

💡 **Goal:** Write structured, reusable, and modular code.



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Data Handling

- **NumPy:**
 - Arrays, vectorization, broadcasting, math operations
- **Pandas:**
 - Series, DataFrames, merging, groupby, pivot, missing values
- **Matplotlib & Seaborn:**
 - Data visualization and styling plots
- **Data Cleaning:**
 - Handling nulls,
 - duplicates,
 - outliers
- **Exploratory Data Analysis (EDA):**
 - Summary stats, correlation, trends
 -

💡 **Goal:** Work confidently with real-world datasets.



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Advanced Python for Data Science

- **Iterators & Generators:** Memory-efficient loops
- **Decorators:** Function enhancements
- **Context Managers:** “with” statements for clean code
- **File Handling:**
 - CSV,
 - Excel,
 - JSON,
 - XML
- **APIs & Web Scraping:** Using requests, BeautifulSoup, json
- **Virtual Environments:** venv, pipenv for dependency management

💡 **Goal:** Write production-grade, efficient, and scalable Python code.



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Data Science & Machine Learning

- **Libraries:** Scikit-learn, Statsmodels, XGBoost
- **ML Basics:** Regression, Classification, Clustering
- **Feature Engineering:** Encoding, scaling, feature selection
- **Model Evaluation:** Cross-validation, confusion matrix, metrics
- **Visualization:** Plot model performance, ROC curves
- **Deep Learning (Intro):** Basics of TensorFlow and PyTorch

💡 **Goal:** Build, train, and evaluate end-to-end ML models.



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Statistics & Mathematics

- **Descriptive Stats:** Mean, Median, Mode, Variance, Std Dev
- **Probability:** Distributions, Bayes' Theorem, Random Variables
- **Hypothesis Testing:**
 - t-test,
 - chi-square,
 - ANOVA
- **Linear Algebra:** Matrices, Vectors, Eigenvalues
- **Calculus (Applied):** Derivatives in gradient descent

💡 **Goal:** Build a mathematical foundation for machine learning.



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Data Engineering Essentials

- **SQL Integration:** Fetching data into Python
- **ETL Pipelines:** Extract, Transform, Load automation using Python
- **Working with Big Data:** Intro to PySpark & Dask
- **APIs & JSON:** Connecting and parsing APIs
- **Scheduling Jobs:** Using cron or Airflow basics

💡 **Goal:** Handle large-scale data pipelines efficiently.



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Automation & Scripting

- **Web Automation:** Selenium, Playwright
- **Email & File Automation:** smtplib, os, shutil, pathlib
- **Task Scheduling:** Automating repetitive data tasks
- **Logging:** Logging module for debugging

💡 **Goal:** Use Python to automate daily business & data tasks.



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Projects to Build

- E-commerce Sales Prediction
- Movie Recommendation System
- Stock Market Analysis using APIs
- Data Cleaning Automation Script
- Resume Screening using NLP
- Customer Churn Prediction

💡 **Goal:** Create a practical portfolio for interviews and freelancing.



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Learning Platforms

- **freeCodeCamp** – Python + ML full courses
- **Kaggle** – Datasets + Competitions
- **LeetCode / HackerRank** – Python coding practice
- **DataCamp / Coursera** – Structured learning paths
- **YouTube** – Krish Naik, Corey Schafer, freeCodeCamp



LeetCode



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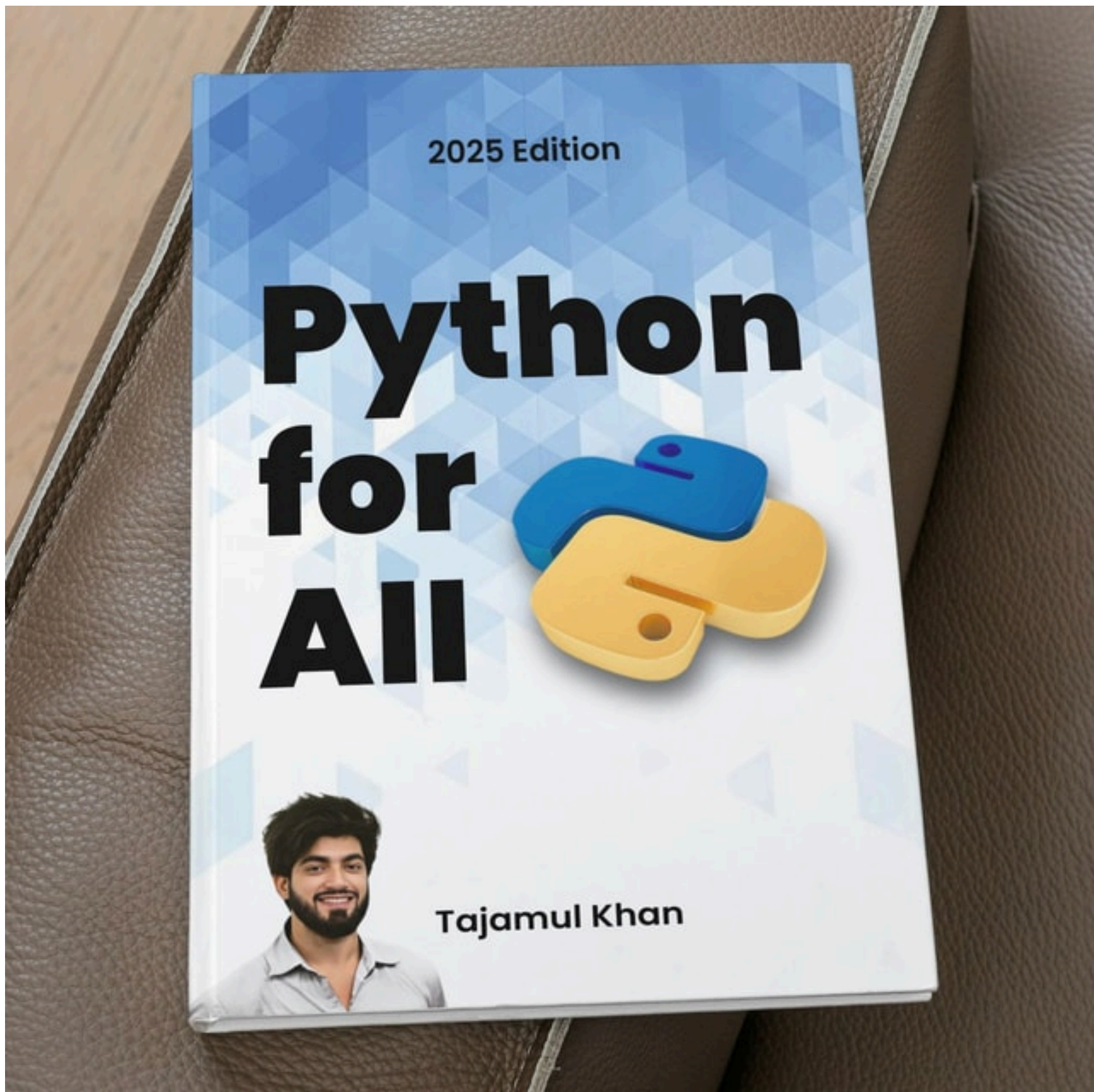


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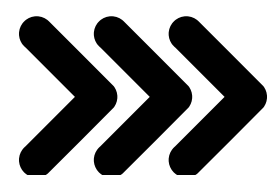
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